DO NOW — RECALL

Insertion sort: trace and implement

Name: _____ Class: _____ Date: _____

1. What is [5,2,4] after one complete ascending bubble-sort pass?

2. What is the value of i after i++ when i was 2?

3. Is a stack FIFO or LIFO?

4. What change makes bubble sort terminate early on sorted data?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

[2,4,5].

2. Model answer

3.

3. Model answer

LIFO: last in, first out.

4. Model answer

Track whether any swap occurred; if none occurred during a complete pass, stop.

Vocabulary — know these terms

key — Value being inserted into the sorted prefix.

sorted prefix — Initial segment already placed in order.

shift — Move a value one index to create space.

in-place — Uses little additional data storage.

stable — Equal elements can preserve relative order.

NEW LEARNING

Read, annotate and explain

One key at a time

The prefix to the left of i is already sorted; take a[i] as the key.

Make room

Shift larger values in the sorted prefix one position to the right.

Insert the key

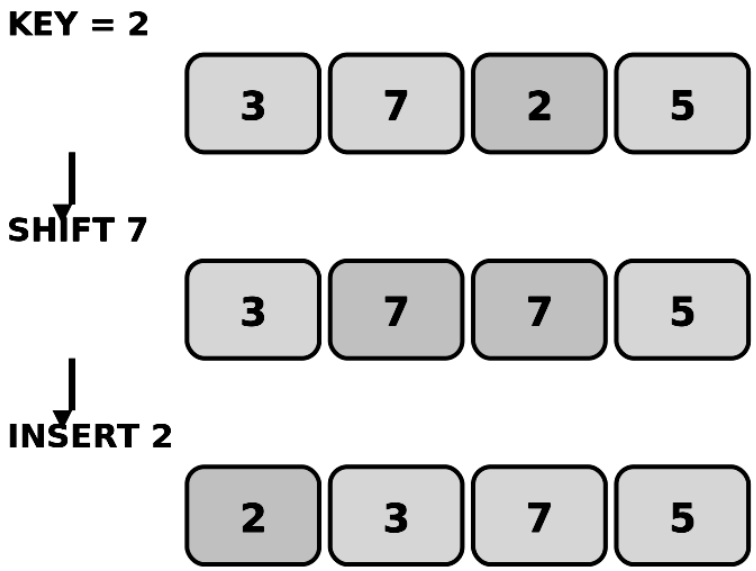
Place it at j+1; the sorted prefix grows by one element.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read the code; predict its effect



Worked algorithm states: KEY = 2 3,7,2,5; SHIFT 7 3,7,7,5; INSERT 2 2,3,7,5

I DO — WORKED EXAMPLE

Follow the worked method

Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].

Predict first

Identify the shifts needed.

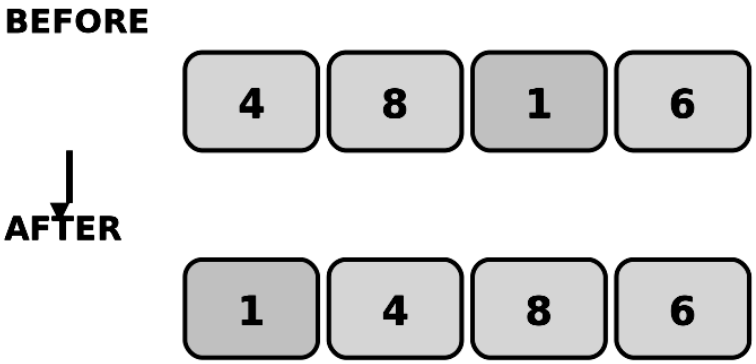
Model solution

Shift 8 then 4 to the right and insert 1 at index 0 → [1,4,8,6].

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].



Worked algorithm states: BEFORE 4,8,1,6; AFTER 1,4,8,6

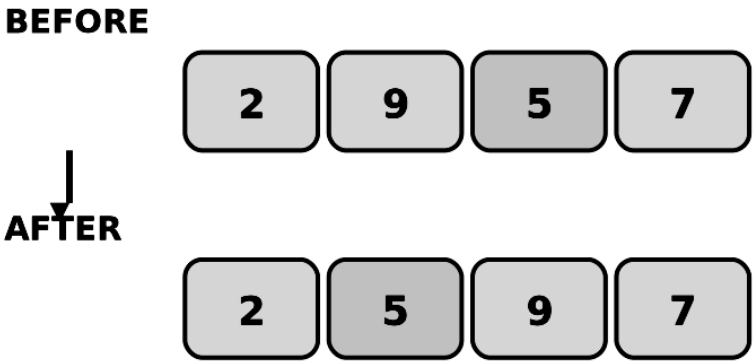
WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert key 5 into the sorted prefix [2,9] of [2,9,5,7].



Worked algorithm states: BEFORE 2,9,5,7; AFTER 2,5,9,7

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Which value must move first?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Insert key 3 into [2,6,3,8], assuming [2,6] is sorted.

Question 2

Trace insertion sort fully on [5,1,4].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why insertion sort is efficient on almost-sorted input.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Insertion sort always swaps each element with its neighbour.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model solution

9 moves one index right; 2 stays. Insert 5 at index 1 → [2,5,9,7].

IMPROVE: Explain one correction to your own reasoning.